

DP Computer Science: A3.2.7 Denormalising Databases

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Explain the concept of **database denormalisation**
 - Discuss the **advantages** of denormalising a database with examples
 - Discuss the **disadvantages** of denormalising a database with examples
 - Identify scenarios, particularly **read-intensive applications**, where denormalisation might be beneficial
 - **Evaluate** the trade-off between simplified query structures and the risks of data redundancy when considering denormalisation
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Key Vocabulary

Term	Definition
Denormalisation	The process of intentionally introducing redundancy into a database to improve query performance
Normalisation	The process of organising data to reduce redundancy and improve integrity
Read-intensive application	An application where data is read much more frequently than it is written (e.g., reports, dashboards)
Write-intensive application	An application where data is written frequently (e.g., transaction processing)
JOIN	An SQL operation that combines data from multiple tables

Term	Definition
Redundancy	The duplication of data across multiple places in a database
Data integrity	The accuracy and consistency of data in a database
Denormalisation trade-off	The balance between improved read performance and increased data management complexity

IB Approaches to Learning (ATL)

Skill	Description
Thinking (Critical Thinking)	Analysing the advantages and disadvantages of denormalisation to form reasoned evaluations
Thinking (Transfer)	Applying understanding of normalisation to the concept of denormalisation in different contexts
Communication	Articulating understanding of denormalisation, its pros and cons, and when it might be appropriate
Research	Researching further examples of denormalisation in real-world systems

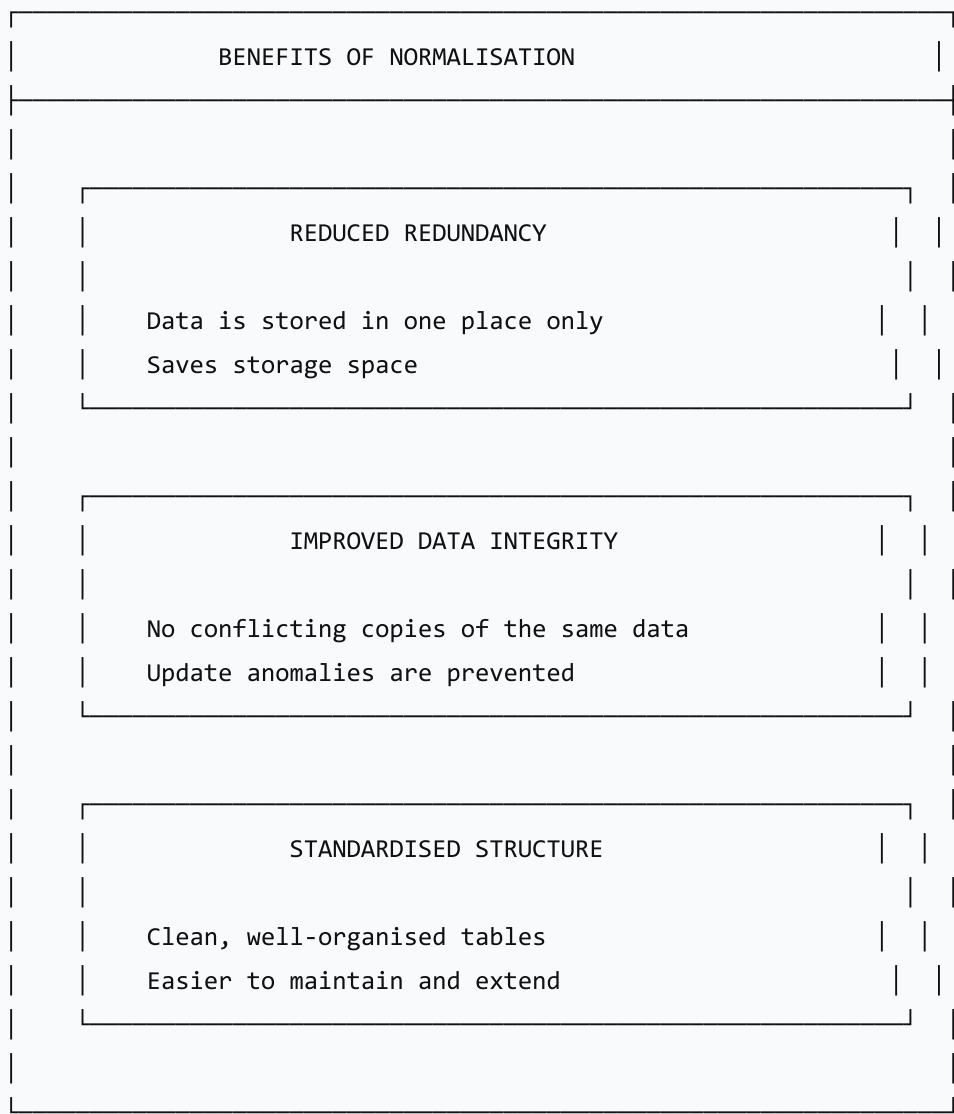
2. Recap: Database Normalisation

What is Normalisation?

Normalisation is the process of organising data to reduce redundancy and improve data integrity. It ensures that data is stored efficiently and consistently.

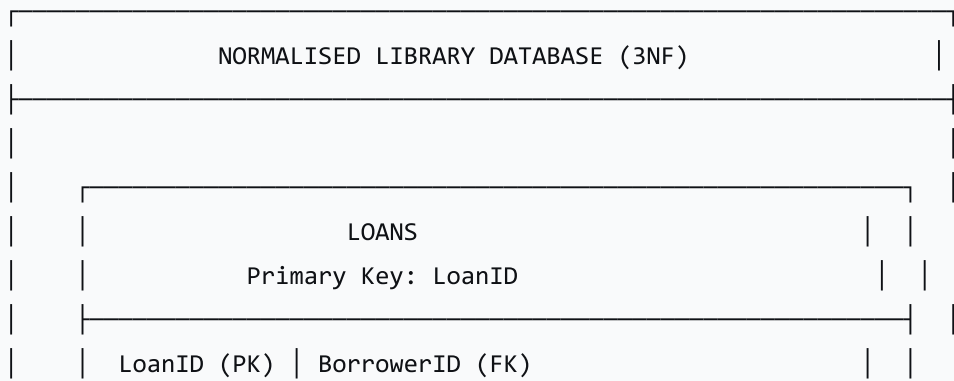
Benefits of Normalisation

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Normalised Library Database (3NF)

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1	101
2	102

BORROWERS		
Primary Key: BorrowerID		
BorrowerID (PK)	BorrowerName	BorrowerAddress
101	Alice Smith	1 Main St
102	Bob Jones	2 Oak Ave

LOAN_ITEMS	
Primary Key: (LoanID, BookTitle)	
LoanID (PK)(FK)	BookTitle (PK)(FK)
1	SQL Guide
1	Python
2	Data Structures

To retrieve a borrower's loan details with book titles:
Need to JOIN Loans, Borrowers, and LoanItems

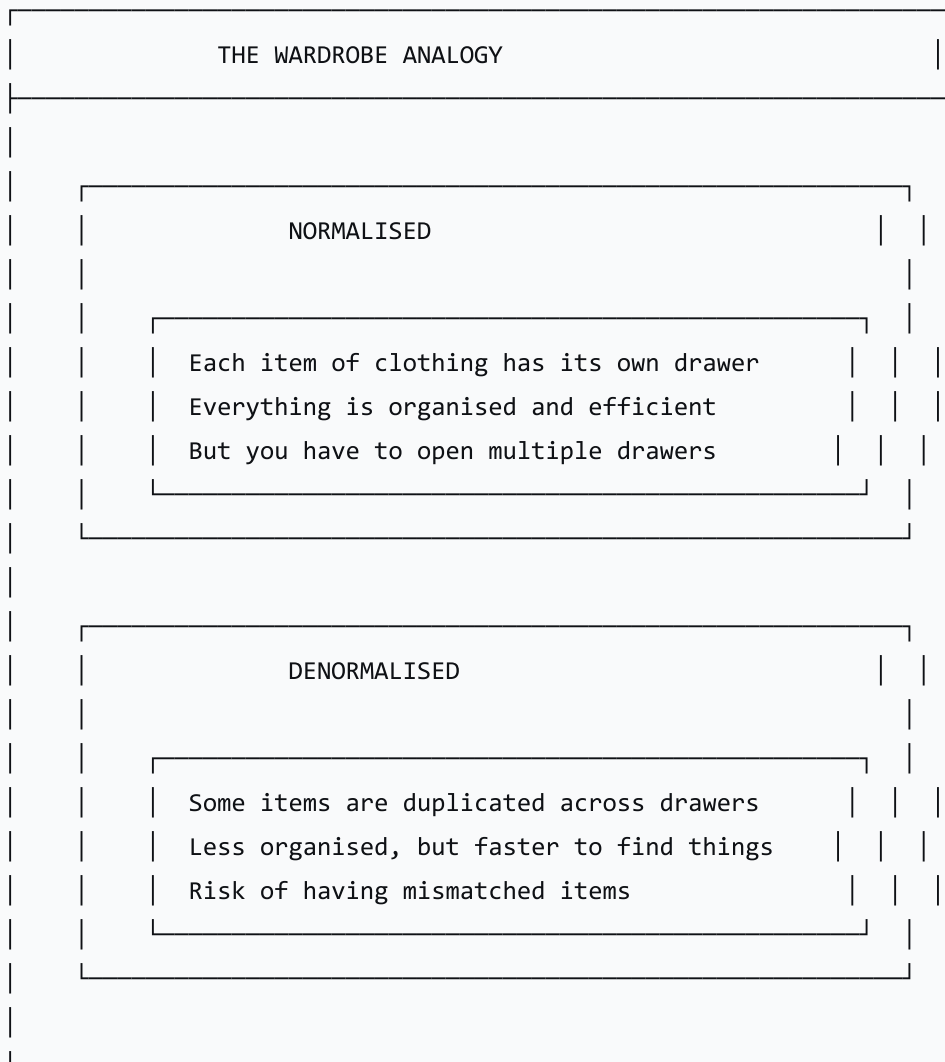
3. Introducing Database Denormalisation

What is Denormalisation?

Denormalisation is the process of intentionally introducing redundancy into a database to improve query performance, particularly for read-heavy applications.

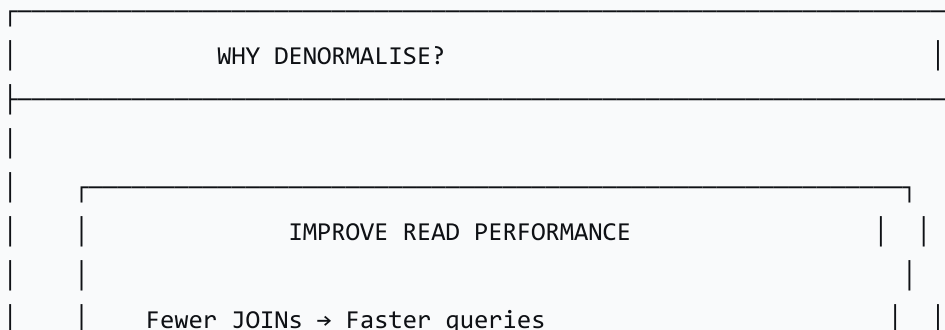
The Wardrobe Analogy

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Why Denormalise?

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Reduced complexity in queries

SIMPLIFY QUERY STRUCTURE

Less complex queries

Single table lookups for common data

OPTIMISE FOR READ-HEAVY APPS

Applications where reads outweigh writes

Reports, dashboards, analytics

4. Denormalisation Example

Normalised Query (Multiple JOINS)

sql

-- Query to get Loan details with borrower name and book title

SELECT

B.BorrowerName,
B.BorrowerAddress,
L.LoanID,
LI.BookTitle

FROM Loans L

JOIN Borrowers B **ON** L.BorrowerID = B.BorrowerID

JOIN LoanItems LI **ON** L.LoanID = LI.LoanID

WHERE L.LoanID = 1;

Denormalised Table (Single Table Lookup)

text

LOAN_DETAILS				
(Denormalised for performance)				
LoanID	BorrowerName	BorrowerAddr	BookTitle	
1	Alice Smith	1 Main St	SQL Guide	
1	Alice Smith	1 Main St	Python	
2	Bob Jones	2 Oak Ave	Data Structures	

- ✓ One table → No JOINS needed
- ✓ Faster read performance
- ✗ Redundant data (borrower info repeated)
- ✗ Update anomalies possible

Query Comparison

Normalised Query	Denormalised Query
Multiple tables	Single table
JOIN operations	No JOINS
Slower for large datasets	Faster for large datasets
Complexity increases	Simpler structure
Better data integrity	Potential redundancy issues

5. Advantages of Denormalisation

Advantage	Description	Example
Faster Read Performance	Fewer JOINS = faster queries	Dashboard reports load instantly
Simpler Queries	Single table lookups are easier to write	Query for daily sales report
Reduced Query Complexity	Less complex SQL statements	Analysts can query more easily
Better Performance for Aggregations	Pre-calculated totals and counts	Monthly sales totals
Suitable for Read-Heavy Applications	Optimised for reporting and analytics	Business intelligence dashboards

When to Consider Denormalisation

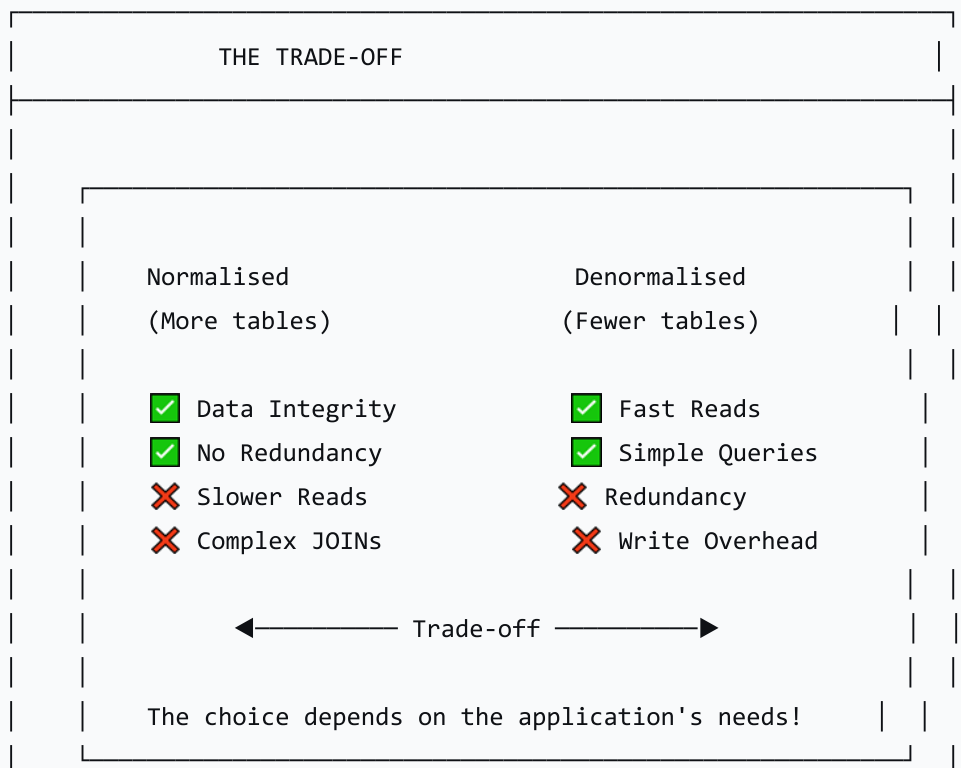
Scenario	Why Denormalise?
Reporting/Dashboards	Frequent reads of aggregated data
Data Warehousing	OLAP systems benefit from denormalisation
Read-Only Applications	Data is rarely modified
Performance-Critical Systems	Need fast response times
Caching	Pre-calculated results for quick access

6. Disadvantages of Denormalisation

Disadvantage	Description	Example
Data Redundancy	Duplicate data across tables	Borrowers' addresses stored in multiple places
Update Anomalies	Must update data in multiple places	Change address → must update all copies
Increased Storage Space	Redundant data consumes more space	Larger database storage requirements
Decreased Write Performance	Updates become slower	More data to update for each write operation
Increased Complexity of Updates	More complex update logic	Need to ensure all copies remain consistent

The Denormalisation Trade-Off

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7. Denormalisation vs. Normalisation

Aspect	Normalisation	Denormalisation
Data Redundancy	Minimal	High
Data Integrity	Excellent	Requires careful management
Read Performance	Slower (JOINS required)	Faster (fewer JOINS)
Write Performance	Faster (single update)	Slower (multiple updates)
Storage Space	Efficient	More space required
Query Complexity	Complex JOINS	Simple SELECTs
Maintenance	Easier (one source of truth)	Harder (manage redundancy)
Best For	OLTP (transaction processing)	OLAP (analytical/reporting)

8. Student Activity: Evaluation Scenario

Scenario

An e-commerce company has a normalised database for their order management system. They are experiencing slow dashboard performance when generating daily sales reports. The reports require joining the Orders, Customers, Products, and OrderItems tables.

Task

Evaluate whether denormalisation is appropriate for this scenario. Consider:

- 1. What is the problem?
- 2. What are the trade-offs?
- 3. Is denormalisation the right solution?
- 4. What alternative solutions exist?

Evaluation Framework

Factor	Consideration
Read/Write Ratio	Reports are read-heavy; orders are write-heavy
Performance Impact	Reports are slow; denormalisation would speed them up
Redundancy Risk	Adding redundant data could cause inconsistency
Update Complexity	If orders change, multiple places must be updated
Alternative Solutions	Caching, data warehousing, materialised views

Sample Evaluation

"Denormalisation could be beneficial for this e-commerce scenario because the reports are read-heavy and require complex JOINS. However, careful consideration must be given to maintaining data consistency when orders are updated. A potential solution could be to create a denormalised reporting table that is updated periodically (e.g., nightly) rather than denormalising the transactional tables directly."

9. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction	5 min	Learning objectives; connection to ATL skills

Phase	Time	Activity
Recap: Normalisation	5 min	Briefly review normalisation and its benefits
Introducing Denormalisation	5 min	Definition; wardrobe analogy
Advantages & Disadvantages	10 min	Detailed explanations with examples
Example Walkthrough	15 min	Normalised vs. denormalised library database
Student Activity	15 min	E-commerce scenario evaluation
Wrap-up	5 min	Summary; Q&A

10. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Student Activity	Evaluate ability to assess denormalisation scenarios
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is database denormalisation?
2. Why would you denormalise a database?
3. What is a disadvantage of denormalisation?
4. When is denormalisation most beneficial?

11. Differentiation

Level	Strategy
Support	Provide clear examples and visual diagrams
Challenge	Research real-world denormalisation examples; explore data warehousing concepts

12. Resources

- **Presentation** (Slide Deck)
- **eBook** (A3.2.7 with examples)
- **Quizlet** (A3.2.7 vocabulary flashcards)
- **Activity:** Evaluation scenario

13. Summary: Key Takeaways

Concept	Key Point
Denormalisation	Intentional introduction of redundancy to improve read performance
Normalisation	Reduces redundancy; improves integrity
Advantages	Faster reads, simpler queries
Disadvantages	Redundancy, update anomalies, write overhead
Trade-off	Read performance vs. data integrity
Best For	Read-heavy applications (reports, dashboards)

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KEY REMINDERS

- ✓ Denormalisation = intentional redundancy
- ✓ Purpose = improve read performance
- ✓ Advantages = faster reads, simpler queries
- ✓ Disadvantages = redundancy, update issues
- ✓ Trade-off = performance vs. integrity
- ✓ Best for = read-heavy applications

"Denormalisation is a deliberate trade-off—sacrificing some data integrity for significant gains in read performance."